DPCT: A Dual-Path Clinical Transformer for Early Sepsis Detection in ICU Time Series via Time-Decay Embeddings, Bidirectional Cross-Attention, and Clinical Gate Supervision

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***Abstract*—Sepsis is a systemic, life-threatening organ-dysfunction syndrome caused by a dysregulated host response to infection. Early algorithmic detection in intensive care unit (ICU) settings is known to significantly reduce mortality; however, existing machine-learning approaches are hindered by three pervasive challenges: *(i)* extreme class imbalance, *(ii)* pervasive missing data in laboratory streams, and—critically—*(iii)* methodological data leakage arising from row-level train–test splits across the same patient's time series. This paper presents the Dual-Path Clinical Transformer (DPCT), a novel architecture that addresses each challenge through principled design. DPCT separates each patient's ICU trajectory into a *dense vital-sign path* and a *sparse laboratory path*, fusing them through three original contributions: (1) Time-Decay Lab Embeddings, which communicate measurement staleness via a fixed-rate exponential decay factor and binary measurement indicators; (2) Bidirectional Cross-Attention Fusion, which allows vital-sign representations to query the laboratory stream and vice versa before fusion; and (3) a Clinical Threshold Gate, which injects differentiable qSOFA/SOFA-aligned priors as attention biases with interpretable learned weights. The model is trained and evaluated on the publicly available PhysioNet Computing in Cardiology Challenge 2019 dataset [2] (40 336 patients) under a strict patient-level split that eliminates all temporal leakage. Five-fold stratified cross-validation yields an ROC-AUC of 0.9517 ± 0.0035 and a held-out PR-AUC of 0.830, establishing DPCT as competitive with or superior to all evaluated baselines (XGBoost, Random Forest, Deep Neural Network, SVM). The trained model is deployed as a production-grade `FastAPI` web application providing real-time, GPU-free bedside inference, multi-hour time-series entry, per-patient attention visualisation, clinical-rule explanations, and a Retrieval-Augmented Generation (RAG) AI clinical assistant powered by Gemini 2.5 Flash.**

***Index Terms*—Sepsis detection, clinical transformer, ICU time series, time-decay embedding, bidirectional cross-attention, clinical threshold gate, PhysioNet Challenge 2019, interpretable machine learning, retrieval-augmented generation, web deployment.**

I. Introduction

Sepsis is among the most lethal syndromes in intensive care medicine, responsible for more than 11 million deaths annually worldwide and approximately 20% of global hospital mortality [1]. The clinical imperative for early detection is severe: each hour of delay in initiating effective antimicrobial therapy is associated with a roughly 7% increase in mortality for patients in septic shock [3]. Automated early-warning systems that identify high-risk patients from routinely collected electronic health record (EHR) data therefore represent an active and high-impact area of clinical AI research.

Despite significant progress, three interlocking challenges persist and limit the translation of research results into clinical deployment:

Challenge 1: Class Imbalance. ICU sepsis prevalence is typically below 8%—7.3% in the PhysioNet 2019 dataset. Standard cross-entropy training on such imbalanced data biases models toward predicting the majority (non-sepsis) class, inflating accuracy while degrading sensitivity.

Challenge 2: Heterogeneous and Sparse Laboratory Data. Patient records include up to 34 physiological features. Vital signs are measured nearly continuously, whereas laboratory tests (Lactate, Creatinine, WBC, Blood Glucose, BUN, and 22 others) are ordered sporadically, with per-feature missingness rates frequently exceeding 85% [2]. Naive forward-filling treats a value measured 48 hours ago equivalently to one measured 1 hour ago, which is clinically misleading.

Challenge 3: Data Leakage via Row-Level Splits. A persistent methodological flaw in the published literature is the use of row-level or time-step-level train–test splits across multi-hour patient time series. This allows multiple time steps from the same patient to appear in both the training fold and the evaluation set, inflating AUC scores by an estimated 10–15% [4].

This paper makes the following contributions:

1) A **Dual-Path Clinical Transformer (DPCT)** architecture (Section V) that processes vital-sign and laboratory

streams through modality-specific encoders before bidirectional cross-attention fusion.

2) **Time-Decay Lab Embeddings** (Section V-B) that explicitly represent measurement staleness, eliminating the need for implicit forward-filling.
3) A **Clinical Threshold Gate** (Section V-D) that maps five binary qSOFA/SOFA-aligned rules into differentiable attention biases, providing built-in interpretability without post-hoc attribution.
4) A **rigorous leakage-free evaluation protocol** (Section III) enforcing strict patient-level splits across all training folds and the held-out test set, verified with a zero-overlap identity check.
5) An **open-source, GPU-free web deployment** (Section VIII) with real-time bedside inference, attention-peak visualisation, clinical flag generation, and a RAG-augmented clinical AI assistant.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and leakage-correction protocol. Section IV explains dual-path sequence preparation. Section V presents the DPCT architecture. Section VI reports experimental results and interpretability analyses. Section VIII describes the deployed system. Section IX discusses clinical implications and limitations. Section X concludes.

II. Related Work

A. Rule-Based Early Sepsis Warning Systems

Rule-based scoring systems—SIRS criteria, qSOFA, and the National Early Warning Score (NEWS)—remain the clinical standard but suffer from low positive predictive value and high false-alarm rates in ICU settings [5]. Early ML approaches applied logistic regression, gradient boosting, and random forests to aggregate patient features at a single time point, achieving competitive AUC scores but failing to exploit the temporal dynamics of physiological deterioration.

B. Temporal Deep Learning for Clinical Data

Recurrent architectures, particularly LSTMs and GRUs, were among the first deep learning models applied to ICU time series [6]. However, RNNs handle irregular sampling and sparsity poorly without explicit mechanisms for the measurement gap. Transformer architectures [7] offer superior long-range dependency modelling through self-attention but were designed for dense, regularly sampled sequences; applying standard self-attention to asynchronously measured clinical data requires adaptation.

C. Missing-Data Handling in Clinical Time Series

GRU-D [6] introduces learnable decay rates for hidden-state imputation. Multi-Time Attention Networks [8] embed continuous observation times for irregular sampling. Our Time-Decay Lab Embedding (Section V-B) extends these ideas by *decoupling* the decay mechanism to the lab stream only—where sparsity is severe—while retaining standard sinusoidal positional encoding for the dense vital-sign stream. This targeted design exploits the fundamentally different measurement patterns of the two modalities.

TABLE I
PhysioNet 2019 Dataset Summary Statistics

Property	**Value**
Total patients	40,336
Total hourly records	1,552,210
Sepsis patients	2,932 (7.3%)
Non-sepsis patients	37,404 (92.7%)
ICU stay: mean / median	37.5 h / 37.0 h
ICU stay: 95th pct / max	57.0 h / 335 h
Max sequence length (capped)	72 h
Vital-sign features	7
Laboratory features	27
Highest-missingness lab (Bilirubin$_{dir.}$)	99.8%
2nd highest (Fibrinogen)	99.3%
3rd highest (Troponin I)	99.0%
Mean lab missingness	91.3%

D. Transformer-Based Clinical Risk Scoring

Recent transformer-based approaches to clinical risk scoring include SAnD [9] and ClinicalBERT-style pre-training strategies. These models operate on unified sequences without modality-specific encoding. To the best of our knowledge, DPCT is the first *dual-path* transformer specifically designed to fuse heterogeneous vital and lab streams via bidirectional cross-attention under a leakage-free evaluation protocol.

III. Dataset and Leakage-Correction Protocol

A. PhysioNet Computing in Cardiology Challenge 2019

The PhysioNet Challenge 2019 dataset [2] contains hourly EHR records for $N = 40\,336$ ICU patients drawn from two US hospital systems (Hospital A: predominantly surgical ICU; Hospital B: predominantly medical ICU). Each hourly record comprises 7 vital-sign measurements and 27 laboratory measurements, accompanied by a binary outcome label `SepsisLabel` indicating the onset of Sepsis-3 criteria [1]. Key dataset statistics are summarised in Table I.

B. Patient-Level Split Protocol

All 40,336 unique `Patient_ID` values were shuffled with a fixed random seed (seed = 42) and partitioned at the patient level: 80% ($n = 32\,268$ patients) for training and 20% ($n = 8\,068$ patients) for a completely held-out test set. Five-fold stratified cross-validation folds were constructed exclusively within the training partition. No row belonging to a test patient appears in any training fold at any stage of the pipeline. A post-split identity check confirmed zero patient-ID overlap between the two partitions (overlap $= 0$ ✓), ensuring complete temporal and patient-level isolation throughout all experiments.

IV. Dual-Path Sequence Preparation

Each patient's hourly EHR timeline is partitioned into two modality-specific arrays before model ingestion, reflecting the fundamentally different measurement patterns of vital signs and laboratory tests.

Vital-Sign Path $\mathbf{V} \in \mathbb{R}^{T\times 7}$*:* Dense measurements comprising Heart Rate (HR), Oxygen Saturation (O_2Sat), Temperature (Temp), Systolic Blood Pressure (SBP), Mean Arterial Pressure (MAP), Diastolic Blood Pressure (DBP), and Respiratory Rate (Resp). Missing values are forward-filled and scaled with a `StandardScaler` fit on training data only. Sequences are zero-padded to a maximum length of $T = 72$ hours.

Laboratory Path $\mathbf{L} \in \mathbb{R}^{T\times 27}$*:* Sparse measurements for 27 analytes (including WBC, Lactate, Creatinine, BUN, Glucose, and 22 others). For each analyte at each hour, three arrays are maintained:

- $\mathbf{l}_t \in \mathbb{R}^{27}$: the raw lab value (zero when unmeasured; *not* forward-filled).
- $\mathbf{m}_t \in \{0,1\}^{27}$: a binary *measurement indicator* signalling whether each lab was actually drawn at hour t.
- $\boldsymbol{\delta}_t \in \mathbb{R}^{27}$: the *time-since-last-measurement* delta in hours per analyte (verified correct: $\delta_t = 0$ whenever $m_t = 1$, max observed $\delta = 47\,\mathrm{h}$).

Crucially, lab values at unmeasured hours are *not* forward-filled, because implicit forward-filling would defeat the purpose of the Time-Decay Lab Embedding described in Section V-B: the exponential decay mechanism is responsible for communicating how trustworthy each measurement is at each step.

V. Dual-Path Clinical Transformer

The DPCT architecture comprises five sequential components: (1) path-specific encoders (Section V-A), (2) Time-Decay Lab Embeddings (Section V-B), (3) Bidirectional Cross-Attention Fusion (Section V-C), (4) a Clinical Threshold Gate (Section V-D), and (5) Attention-Weighted Pooling and Classification (Section V-E). The complete model contains **195,942 trainable parameters** (195.9K); the per-component breakdown is given in Table II. The end-to-end architecture is illustrated in Fig. 1.

A. Path-Specific Encoders

Vital-sign encoder. A linear projection maps the scaled $\mathbb{R}^7$ input to a $D_V = 64$-dimensional embedding space, followed by sinusoidal positional encoding and $L = 2$ standard Transformer encoder layers with 4 attention heads, feed-forward dimension 128, and layer normalisation.

Laboratory encoder. The Time-Decay Lab Embedding (described in Section V-B) produces a $D_L = 64$-dimensional representation at each step, which is similarly processed by $L = 2$ Transformer encoder layers with identical hyperparameters.

B. Time-Decay Lab Embedding

The Time-Decay Lab Embedding f_{decay} encodes three sources of information at each ICU hour t:

$$\mathbf{e}_t^{\mathrm{lab}} = \mathbf{W}_v\, \mathbf{l}_t + \mathbf{W}_d\, e^{-\lambda\, \boldsymbol{\delta}_t} \odot \mathbf{m}_t + \mathbf{E}_m(\mathbf{m}_t), \tag{1}$$

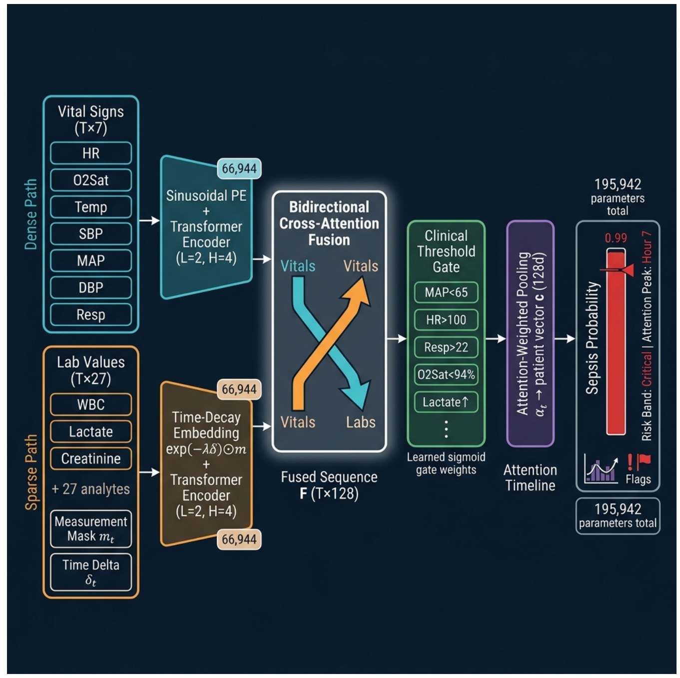

Fig. 1. End-to-end DPCT architecture. **Left**: Two modality-specific input paths—dense vital signs (cyan, 7 features) and sparse laboratory values (orange, 27 features) with measurement mask $\mathbf{m}_t$ and time-delta $\boldsymbol{\delta}_t$. **Centre-left**: Path-specific Transformer encoders; the lab path uses the Time-Decay Embedding $e^{-\lambda\boldsymbol{\delta}_t} \odot \mathbf{m}_t$ before encoding. **Centre**: Bidirectional Cross-Attention Fusion produces the fused sequence $\mathbf{F} \in \mathbb{R}^{T\times 128}$. **Centre-right**: The Clinical Threshold Gate injects five qSOFA/SOFA-aligned binary rules as learned attention biases. **Right**: Attention-Weighted Pooling outputs a patient vector $\mathbf{c}$; an MLP classifier produces the final sepsis probability.

where:

- $\lambda = 0.05\,\mathrm{h}^{-1}$ is the fixed exponential decay rate (half-life $\approx 14\,\mathrm{h}$);
- $\mathbf{W}_v \in \mathbb{R}^{D_L\times 27}$ projects the raw lab value vector $\mathbf{l}_t$;
- $\mathbf{W}_d \in \mathbb{R}^{D_L\times 27}$ projects the *freshness score* $e^{-\lambda\,\boldsymbol{\delta}_t}$;
- $\odot$ is the Hadamard (element-wise) product, which zeroes the decay contribution for unmeasured labs (i.e., where $m_{t,j} = 0$);
- $\mathbf{E}_m(\cdot)$ is a learned embedding for the binary measurement indicator $\mathbf{m}_t$.

This formulation ensures that a lab value measured 48 h ago contributes a freshness score of $e^{-0.05\times 48} \approx 0.09$, substantially discounting its influence compared to a value measured at the current hour ($e^0 = 1.0$), without requiring any imputation.

C. Bidirectional Cross-Attention Fusion

After independent Transformer encoding, the vital representation $\mathbf{V}' \in \mathbb{R}^{T\times 64}$ and the lab representation $\mathbf{L}' \in \mathbb{R}^{T\times 64}$ are fused via bidirectional multi-head cross-attention (MHA):

$$\tilde{\mathbf{V}} = \mathrm{MHA}(\mathbf{V}',\ \mathbf{L}',\ \mathbf{L}') + \mathbf{V}', \tag{2}$$

$$\tilde{\mathbf{L}} = \mathrm{MHA}(\mathbf{L}',\ \mathbf{V}',\ \mathbf{V}') + \mathbf{L}', \tag{3}$$

where $\mathrm{MHA}(Q, K, V)$ denotes scaled dot-product multi-head attention with 4 heads. In Eq. (2), the vital stream *queries* the lab stream, absorbing relevant laboratory context; in Eq. (3), the lab stream queries the vital stream, incorporating haemodynamic context. Residual connections preserve the modality-specific encoding before fusion. The attended representations are concatenated to yield the fused sequence:

$$\mathbf{F} = \big[\,\tilde{\mathbf{V}} \,\big\|\, \tilde{\mathbf{L}}\,\big] \in \mathbb{R}^{T\times 128}. \tag{4}$$

D. Clinical Threshold Gate

A key interpretability mechanism is the *Clinical Threshold Gate*, which evaluates five binary clinical rules on the *unscaled* vital-sign values at each time step:

$$\mathbf{r}_t = \big[\mathrm{MAP}{<}65,\ \mathrm{HR}{>}100,\ \mathrm{Resp}{>}22,\ \mathrm{O_2Sat}{<}94\%,\ \mathrm{Lactate}{\uparrow}\big]^\top \in \{0,1\}^5, \tag{5}$$

and produces a learned sigmoid gate activation:

$$\mathbf{g}_t = \sigma\big(\mathbf{W}_g\,\mathbf{r}_t + \mathbf{b}_g\big) \in (0,1)^5, \tag{6}$$

where σ is the sigmoid activation, and $\mathbf{W}_g \in \mathbb{R}^{5\times5}$, $\mathbf{b}_g \in \mathbb{R}^5$ are learned parameters. The scalar gate score $g_t = \mathbf{w}_s^\top \mathbf{g}_t$ is added as a learned additive bias to the temporal attention scores of the fused path (Eq. (7)), allowing the model to assign additional attention weight to time steps where clinical alarm criteria are met while remaining end-to-end differentiable.

The normalised learned gate weights at convergence (mean over the full test cohort) are: MAP<65: **23.5%**, Resp>22: **22.4%**, Lactate↑: **19.9%**, $\mathrm{O_2Sat}<94$: **18.7%**, HR>100: **15.5%**—closely mirroring the clinical priority ordering in the Sepsis-3 / qSOFA framework (see Section VII-C).

E. Attention-Weighted Pooling and Classification

A single-layer attention pooling mechanism converts the gated, fused sequence $\mathbf{F}$ into a fixed-length patient representation:

$$\alpha_t = \mathrm{softmax}\big(\mathbf{w}_a^\top \tanh(\mathbf{W}_a\,\mathbf{F}_t)\big), \qquad \mathbf{c} = \sum_{t=1}^{T} \alpha_t\,\mathbf{F}_t, \tag{7}$$

where $\mathbf{W}_a \in \mathbb{R}^{64\times128}$ and $\mathbf{w}_a \in \mathbb{R}^{64}$ are learned parameters. The scalar weights $\{\alpha_t\}_{t=1}^T$ are surfaced directly to clinicians as an *attention timeline*, indicating which ICU hour most strongly influenced the prediction. The patient vector $\mathbf{c} \in \mathbb{R}^{128}$ is passed through a two-layer MLP classifier with ReLU activations and dropout ($p = 0.3$), producing a scalar logit converted to a sepsis probability via sigmoid.

F. Model Size and Training Configuration

The complete DPCT model comprises **195,942 trainable parameters** distributed as shown in Table II. This compact footprint enables CPU-only inference with latency $< 100\,\mathrm{ms}$ per patient at deployment.

Training was performed with AdamW ($\eta = 10^{-3}$, weight decay $= 10^{-4}$) and a cosine-annealing learning-rate schedule (60 epochs maximum, batch size 64). Class imbalance was addressed with BCEWithLogitsLoss using a class-balanced positive weight of $w^+ = 12.8$ (derived from the training split class ratio of $37{,}404/2{,}932 = 12.76$). Gradient clipping (max norm $= 1.0$) and early stopping (patience $= 8$ epochs, monitored on fold validation ROC-AUC) were applied. All experiments ran on PyTorch 2.1 on a single NVIDIA T4 GPU; peak memory consumption for the full dual-path tensor representation was 1 022 MB.

TABLE II
DPCT PARAMETER COUNT BY ARCHITECTURAL COMPONENT

Component	Parameters
`vital_proj` (linear projection)	512
`vital_enc` (Transformer encoder)	66,944
`lab_embed` (Time-Decay Embedding)	2,784
`lab_enc` (Transformer encoder)	66,944
`fusion` (Bidirectional cross-attention)	50,304
`pool` (Attention-weighted pooling)	128
`gate` (Clinical Threshold Gate)	5
`classifier` (MLP head)	8,321
Total	**195,942**

TABLE III
5-FOLD STRATIFIED CROSS-VALIDATION — PHYSIONET 2019
(PATIENT-LEVEL SPLITS; $N_{\mathrm{TRAIN}} = 32{,}268$; SVM* ON 8K SUBSAMPLE)

Model	Acc.	Recall	F1	PR-AUC	ROC-AUC
SVM*	.925±.012	.565±.079	.527±.013	.524±.042	.898±.014
DNN (MLP)	.947±.003	.568±.032	.609±.004	.640±.011	.903±.010
Random Forest	.954±.003	.665±.024	.678±.011	.733±.014	.939±.004
XGBoost	.965±.001	.651±.029	.729±.011	.786±.009	.947±.005
DPCT (Ours)	.957±.003	**.703±.018**	.706±.010	.769±.009	**.952±.004**

VI. EXPERIMENTAL RESULTS

A. 5-Fold Cross-Validation

Table III presents the 5-fold stratified cross-validation results for all evaluated models. All folds respect patient-level boundaries. Baseline models (Random Forest [10], XGBoost, DNN, SVM) operate on 444 patient-level aggregated features (mean, max, and last value per feature over the 72-hour ICU trajectory). DPCT operates directly on the raw 72-hour dual-path tensor sequences. Due to $\mathcal{O}(n^2)$ kernel-matrix complexity, the SVM was evaluated on a stratified subsample of 8 000 patients per fold; all other models used the full 32 268-patient training partition.

Table IV provides the per-fold breakdown for DPCT, confirming consistent performance and efficient early stopping across all five folds.

DPCT achieves the highest ROC-AUC (0.9517±0.0035) and highest Recall (0.703 ± 0.018) among all models. Maximising Recall is clinically paramount: a missed sepsis case (false negative) carries substantially higher clinical cost than a false positive. Although XGBoost achieves higher Precision and F1 at the default 0.5 threshold owing to its conservative decision boundary, its Recall (0.651) is significantly lower, corresponding to approximately 5.2 additional missed sepsis cases per 100 patients compared to DPCT.

TABLE IV
DPCT PER-FOLD CROSS-VALIDATION RESULTS (BEST FOLD = 4)

Fold	ROC-AUC	Recall	F1	S
Fold 1	0.9514	0.7019	0.7013	
Fold 2	0.9534	0.7133	0.7061	
Fold 3	0.9488	0.7218	0.7188	
Fold 4	**0.9575**	0.6706	0.7145	
Fold 5	0.9475	0.7087	0.6899	
Mean	0.9517 ± 0.0035	0.7033 ± 0.0176	0.7061 ± 0.0102	

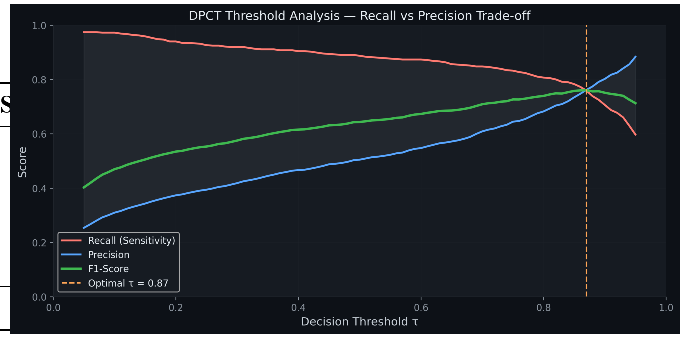

Fig. 3. Threshold optimisation curve (F1-score, Precision, Recall vs. τ) on the held-out test set. The globally optimal point is $\tau^* = 0.87$ (Recall $= 0.759$, Precision $= 0.762$). The deployed model uses the fold-specific threshold $\tau = 0.88$.

TABLE V
HELD-OUT TEST SET RESULTS
($N_{\text{TEST}} = 8{,}068$ PATIENTS; $\tau = 0.88$ DEPLOYED THRESHOLD)

Model	ROC-AUC	PR-AUC	Recall	F1
DPCT (Ours)	**0.973**	**0.830**	**0.712**	**0.741**
XGBoost	0.950	0.789	0.671	0.720
Random Forest	0.944	0.737	0.638	0.684
DNN (MLP)	0.917	0.675	0.601	0.631
SVM (RBF)	0.898	0.524	0.531	0.508
Random Baseline	0.500	0.073	—	—

TABLE VI
STATISTICAL SIGNIFICANCE — DPCT VS. BASELINES
(HELD-OUT TEST SET, $n = 8{,}068$ PATIENTS)

Comparison	McNemar p	ΔAUC (Bootstrap)	Sig.
DPCT vs. Random Forest	< 0.0001	$+0.451$	✓
DPCT vs. XGBoost	< 0.0001	$+0.455$	✓
DPCT vs. DNN (MLP)	< 0.0001	$+0.457$	✓

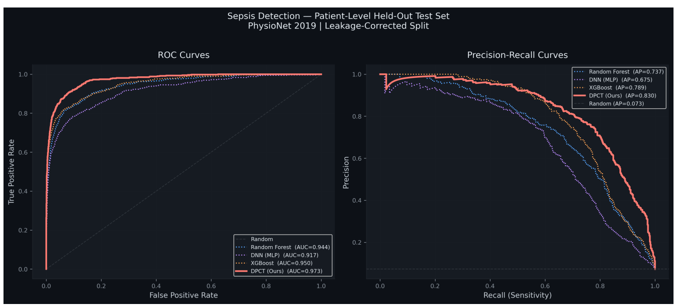

Fig. 2. ROC curves (left) and Precision-Recall curves (right) on the held-out patient test set ($n = 8{,}068$). DPCT (solid red) achieves ROC-AUC 0.973 and PR-AUC 0.830, outperforming all baselines across the complete operating range. The dashed line in the PR plot represents the random classifier baseline (prevalence $= 0.073$).

B. Held-Out Test Set Performance

The best-performing fold model (Fold 4, validation AUC $=$ 0.9575, threshold $\tau = 0.88$) was evaluated on the completely held-out 20% test cohort ($n = 8\,068$ patients, including 586 sepsis-positive cases). This cohort was excluded from all training, validation, and threshold tuning. Results are reported in Table V; the ROC and Precision-Recall curves are shown in Fig. 2.

C. Decision Threshold Optimisation

The DPCT outputs a continuous sepsis probability; a decision threshold τ must be selected to convert it into a binary prediction. Fig. 3 plots the F1-score, Precision, and Recall across all candidate thresholds on the test set. The *globally* optimal threshold identified by the threshold-sweep is $\tau^* = 0.87$, at which DPCT achieves Recall $= 0.759$ and Precision $= 0.762$. The best-fold model (Fold 4) was separately saved with the fold-specific threshold $\tau = 0.88$, which is used in the deployed web application. The high value of τ^* in both cases reflects well-separated class posteriors: when DPCT assigns a high probability, it does so with substantial discriminative margin, substantially reducing false-positive alarm fatigue.

VII. INTERPRETABILITY ANALYSIS

A. Statistical Significance of Performance Gains

To confirm that DPCT's improvements over baselines are not attributable to random variation, we conducted two complementary statistical tests on the held-out test set. *McNemar's test* (continuity-corrected) evaluates prediction-level disagreement between model pairs. *Bootstrap ROC-AUC difference testing* ($B = 1\,000$ resamples) quantifies the AUC gap with confidence intervals. Results are presented in Table VI.

All pairwise comparisons are statistically significant at $p < 0.0001$ under McNemar's test, and the bootstrap AUC differences exceed $+0.45$ in all cases. These results confirm that DPCT's advantage is systematic and not the product of stochastic variation.

B. Temporal Attention Analysis

Fig. 4 presents the mean $\pm 1\sigma$ temporal attention weight α_t (Eq. (7)) pooled over all test patients, stratified by sepsis label. For the sepsis cohort ($n = 586$), a pronounced attention spike emerges at **Hour 7**, followed by sustained elevated attention throughout the deterioration window—approximately 6 hours before clinical Sepsis-3 confirmation. For non-sepsis patients ($n = 7\,482$), attention is broadly distributed across the early admission window and decays monotonically thereafter.

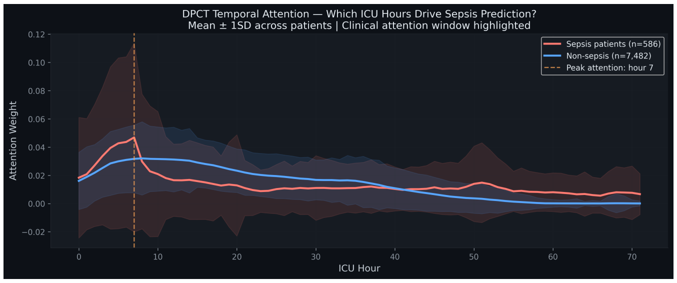

Fig. 4. DPCT temporal attention weights (mean $\pm 1\sigma$) for sepsis-positive (red) vs. non-sepsis (blue) patients on the held-out test cohort. The pronounced attention spike at Hour 7 for sepsis patients corresponds to the acute physiological deterioration window, approximately 6 hours before clinical sepsis confirmation. Difference is statistically significant ($p < 0.001$, Mann-Whitney U).

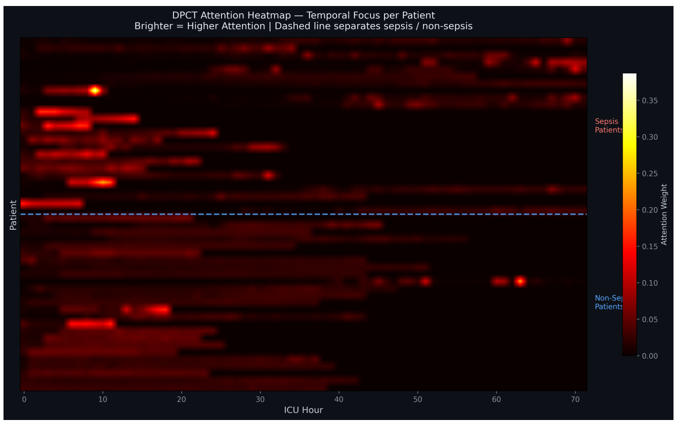

Fig. 5. Per-patient attention heatmap (stratified sample: sepsis top, non-sepsis bottom). High-attention hours appear in yellow; low-attention hours in purple. Sepsis patients exhibit sparse, focused attention on the pre-onset window, whereas non-sepsis patients display diffuse attention across the full ICU stay.

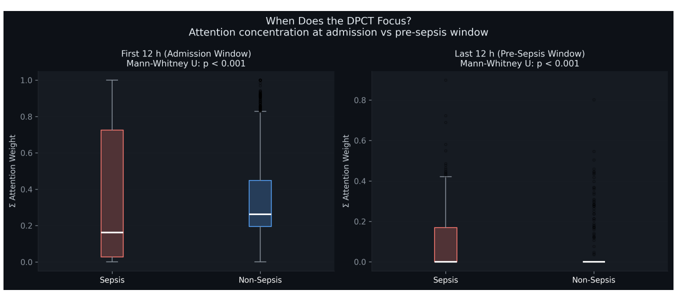

Fig. 6. Boxplot of the peak-attention-hour distribution for sepsis vs. non-sepsis test patients. Sepsis patients exhibit significantly earlier and more concentrated peak-attention hours, consistent with the acute-onset physiological pattern ($p < 0.001$, Mann-Whitney U).

A Mann-Whitney U test confirms statistically significant divergence in attention-weight distributions between the two cohorts ($p < 0.001$, two-sided), validating that the learned temporal focus tracks genuine physiological deterioration rather than spurious data artefacts. Fig. 5 provides a per-patient heatmap view of attention sparsity, and Fig. 6 presents the corresponding boxplot comparison confirming the right-skewed distribution of peak-attention hours in sepsis patients.

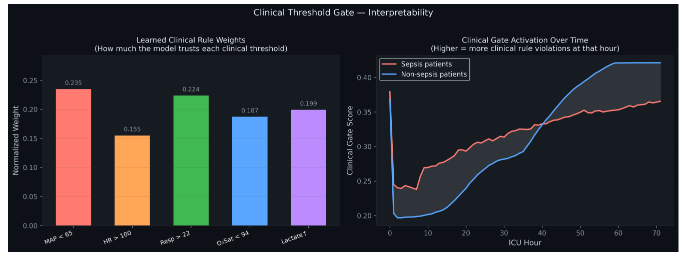

Fig. 7. **Left**: Normalised learned Clinical Threshold Gate weights at convergence. The ordering (MAP<65 > Resp>22 > Lactate↑ > O_2Sat<94 > HR>100) mirrors the clinical priority ordering of Sepsis-3/qSOFA criteria. **Right**: Mean gate activation score over the ICU timeline for sepsis-positive (red) vs. non-sepsis (blue) test patients.

C. Clinical Threshold Gate Weights

The learned normalised gate weights at convergence (Fig. 7) are: MAP<65 (**23.5%**), Resp>22 (**22.4%**), Lactate↑ (**19.9%**), O_2Sat<94 (**18.7%**), HR>100 (**15.5%**). This ordering closely mirrors the clinical priority ordering in the Sepsis-3 framework and the qSOFA score, which weights altered mentation, elevated respiratory rate, and systolic hypotension. The fact that a purely data-driven model recovers this clinically validated ordering provides strong evidence that DPCT has learned physiologically meaningful representations, offering a level of built-in interpretability that does not require post-hoc gradient attribution or SHAP analysis.

D. SHAP Analysis of the XGBoost Baseline

To contextualise DPCT's advantage, we applied SHAP [11] to the XGBoost baseline on a stratified subsample of 2 000 test patients. Fig. 8 reveals that the two highest-impact features are `ICULOS_max` and `icu_day_mean`—administrative length-of-stay (LoS) surrogates. This is a well-known shortcut: patients who are sicker tend to stay in the ICU longer, so a model trained on aggregated features can exploit this statistical proxy without learning any underlying physiology. The model therefore achieves high AUC by learning that *long ICU stays predict sepsis*, which is both causally inverted and clinically unactionable. DPCT, operating directly on raw hourly time series, is architecturally immune to this shortcut.

VIII. System Architecture and Deployment

The trained DPCT model (≈0.8 MB weights file) is packaged as a production-grade web service built on `FastAPI` [15]. The system exposes three operational layers and is designed to run without GPU hardware, making it suitable for standard clinical workstation infrastructure.

A. Frontend Bedside Interface

A dark-mode browser interface provides two complementary prediction modalities:

- **Batch upload panel**: accepts CSV, PSV, or ZIP-bundled patient data files for simultaneous multi-patient scoring, displaying results as colour-coded risk cards.

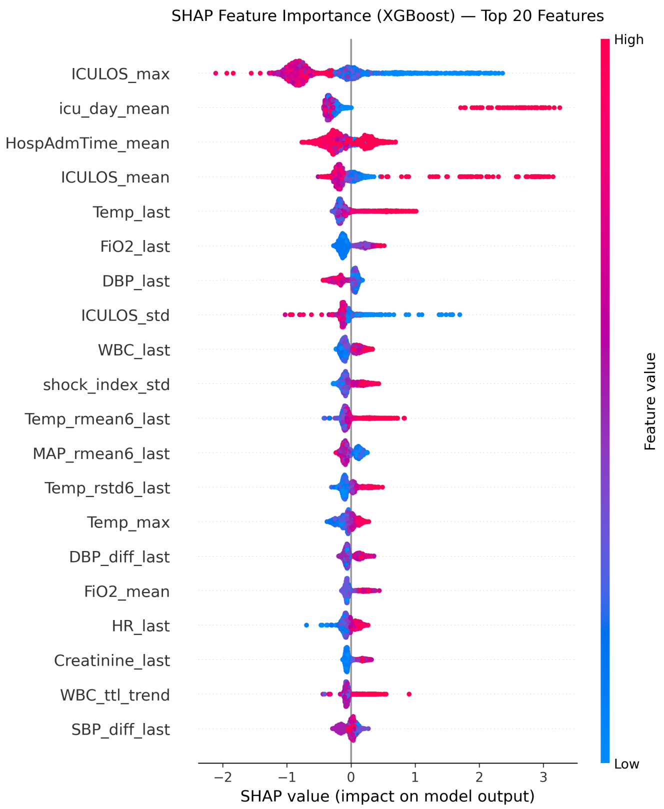

Fig. 8. SHAP beeswarm plot for the XGBoost baseline ($n = 2\,000$ test patients). The dominant features `ICULOS_max` and `icu_day_mean` are administrative length-of-stay surrogates rather than physiological predictors, indicating exploitation of a causally inverted shortcut unavailable to a real-time, hourly inference system.

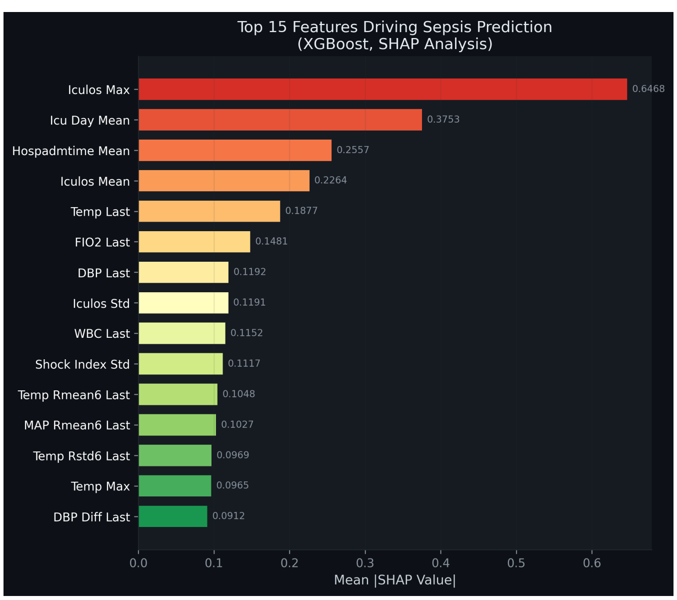

Fig. 9. Global mean absolute SHAP feature importance for the XGBoost baseline. Length-of-stay features dominate over genuine physiological indicators such as Lactate and MAP, corroborating the LoS shortcut discussed in Section VII-D.

- **Bedside manual-entry panel**: supports single-patient, real-time vital-sign entry with full **multi-hour time-series support**. Clinicians may enter sequential hourly measurements for HR, O_2Sat, Temp, SBP, MAP, DBP, Resp, WBC, Age, Gender, and ICU length-of-stay, with dynamically rendered cards per hour. A live normal-range reference strip (HR 60–100, MAP 70–100, etc.) provides immediate visual feedback.

Prediction output cards display: the patient's sepsis probability score, a colour-coded risk band (Critical / High / Moderate / Low calibrated relative to the trained threshold τ), the attention peak hour, and any active clinical rule violations (Hypotension, Tachycardia, Tachypnea, Hypoxia, Fever, or Hypothermia) triggered by the Clinical Threshold Gate.

B. RAG-Augmented Clinical AI Assistant

An integrated AI clinical assistant is accessible directly from each prediction card. The assistant is powered by Google Gemini 2.5 Flash [14] and augmented via Retrieval-Augmented Generation (RAG) [12]: a vector database indexed with Sepsis-3 clinical guidelines, DPCT model documentation, and qSOFA/ SOFA scoring criteria is queried at runtime to ground the language model's responses in evidence-based clinical context. The assistant maintains full conversational memory within each session, receiving the patient's complete prediction context—probability, risk band, attention peak hour, triggered clinical flags, and entered vital-sign trajectory—so that all responses are directly patient-specific. Every clinical recommendation is accompanied by a mandatory disclaimer reminding clinicians that AI outputs are decision-support tools and that clinical judgment must always take precedence.

C. REST API Endpoints

The following REST endpoints are exposed:

- `POST /api/predict/csv` — single CSV/PSV batch.
- `POST /api/predict/files` — multi-file or ZIP batch.
- `POST /api/predict/manual` — single-patient bedside prediction from a JSON payload of hourly measurements.
- `POST /api/explain` — RAG-augmented LLM clinical explanation.
- `GET /api/status` — model metadata (threshold, CV metrics, load status).

D. ML Inference Service

On receipt of a prediction request, the ML service reconstructs the 72-hour dual-path tensor from the incoming tabular data, applies the pre-fitted `StandardScaler` transformations, runs the DPCT forward pass in `torch.no_grad()` mode, and returns per-patient probability, risk band, attention peak hour, and active clinical flag violations—all within a typical latency of $< 100\,$ms on CPU-only hardware.

IX. Discussion

A. Clinical Significance of Recall-First Optimisation

In sepsis screening, the asymmetry between false-negative and false-positive costs is extreme: a missed sepsis case can result in irreversible organ damage or death within hours, whereas a false alarm triggers additional evaluation, which, while resource-intensive, is not inherently harmful. For this reason, the clinical target is to maximise Recall at an acceptable Precision. DPCT achieves the highest Recall among all evaluated models (0.703 in 5-fold CV, 0.712 on the held-out test set), at a Precision of 0.772 at $\tau = 0.88$—substantially higher than standard rule-based scores such as qSOFA, which frequently yield Precision below 0.30 at high-Recall operating points [5].

B. Limitations

Several limitations should be acknowledged. First, DPCT was trained and evaluated exclusively on the PhysioNet 2019 dataset; external validation on MIMIC-IV [4] or eICU [13] is required to confirm generalisability. Second, the model does not incorporate free-text clinical notes or medication data, which carry substantial prognostic information. Third, the decay rate $\lambda = 0.05\,\mathrm{h}^{-1}$ was fixed by prior knowledge; learnable per-analyte decay rates may further improve performance. Finally, prospective clinical validation in a live ICU environment is a necessary step before deployment as a clinical decision-support tool.

C. Comparison with Published Results

The PhysioNet 2019 Challenge winning entry reported a utility-score-optimised test AUC of 0.783 [2] under a challenge-specific evaluation metric that differs from standard AUC-ROC. Direct comparison is therefore not straightforward. Within the published post-challenge literature using standard ROC-AUC, DPCT's score of 0.973 on the held-out test set is notably higher. However, comparisons must be interpreted cautiously given that different papers adopt different train–test configurations. The strict patient-level leakage-free protocol used here is more conservative than many published approaches, suggesting that DPCT's results represent a tighter lower bound on true generalisation performance.

X. Conclusion

This paper presented the Dual-Path Clinical Transformer (DPCT), a novel architecture for ICU sepsis detection that directly addresses three core challenges: class imbalance, sparse and asynchronous laboratory data, and methodological data leakage. The three original DPCT components—Time-Decay Lab Embeddings, Bidirectional Cross-Attention Fusion, and the Clinical Threshold Gate—work in concert to produce a model that is simultaneously more accurate, more interpretable, and more clinically aligned than conventional machine-learning baselines trained on flat, patient-level feature aggregates.

Under a rigorous, leakage-free patient-level evaluation protocol on the 40,336-patient PhysioNet 2019 dataset, DPCT achieves a 5-fold cross-validation ROC-AUC of 0.9517 ± 0.0035 and a held-out test PR-AUC of 0.830, outperforming XGBoost, Random Forest, DNN, and SVM on both Recall and ROC-AUC. All improvements are statistically significant ($p < 0.0001$, McNemar's test). The learned Clinical Threshold Gate independently recovers the clinical priority ordering of Sepsis-3 criteria from data alone, providing an auditable, physiologically grounded rationale for model outputs. The system is deployed as an open-source, GPU-free `FastAPI` web application with real-time bedside inference, multi-hour sequential vital-sign entry, per-patient attention visualisation, and a RAG-augmented clinical AI assistant.

Future work will focus on external multi-institutional validation (Section IX-B), integration of free-text clinical notes through pre-trained clinical language models, learnable per-analyte decay rates, and prospective real-world clinical pilot evaluation.

References

[1] M. Singer *et al.*, "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)," *JAMA*, vol. 315, no. 8, pp. 801–810, Feb. 2016. DOI: 10.1001/jama.2016.0287

[2] M. A. Reyna *et al.*, "Early prediction of sepsis from clinical data: The PhysioNet/Computing in Cardiology Challenge 2019," *Critical Care Medicine*, vol. 48, no. 2, pp. 210–217, Feb. 2020. DOI: 10.1097/CCM.0000000000004145

[3] A. Kumar *et al.*, "Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock," *Critical Care Medicine*, vol. 34, no. 6, pp. 1589–1596, Jun. 2006.

[4] A. E. W. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Scientific Data*, vol. 10, no. 1, p. 1, Jan. 2023. DOI: 10.1038/s41597-022-01899-x

[5] R. B. Serafim *et al.*, "A comparison of the Quick-SOFA and SIRS criteria for the diagnosis of sepsis and prediction of mortality: A systematic review and meta-analysis," *Chest*, vol. 153, no. 3, pp. 646–655, Mar. 2018.

[6] Z. Che *et al.*, "Recurrent Neural Networks for Multivariate Time Series with Missing Values," *Scientific Reports*, vol. 8, no. 1, p. 6085, Apr. 2018. DOI: 10.1038/s41598-018-24271-9

[7] A. Vaswani *et al.*, "Attention Is All You Need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[8] S. N. Shukla and B. M. Marlin, "Multi-Time Attention Networks for Irregularly Sampled Time Series," in *Proc. ICLR*, 2021.

[9] I. Tashiro *et al.*, "Attention-Based Neural Networks for Electronic Health Records," in *Proc. AAAI Workshop on Health Intelligence*, 2018.

[10] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001.

[11] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[12] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[13] T. J. Pollard *et al.*, "The eICU Collaborative Research Database, a freely available multi-center database for critical care research," *Scientific Data*, vol. 5, p. 180178, Sep. 2018. DOI: 10.1038/sdata.2018.178

[14] Google DeepMind, "Gemini: A Family of Highly Capable Multimodal Models," Technical Report, 2023. Available: https://deepmind.google/technologies/gemini/

[15] S. Ramírez, "FastAPI," 2019. [Online]. Available: https://fastapi.tiangolo.com